

## **Yelp Sentiment Analysis: Elevating Customer Experience & Fueling Business Growth**

### **Introduction**

The advent of platforms like Yelp has revolutionized consumer decision-making, enabling individuals to influence their purchasing choices through reviews that evaluate a business's product or service quality. However, Yelp's presentation of an aggregate rating based on all historical reviews can be misleading. This becomes particularly problematic when a business's quality has undergone noticeable changes more recently. Meanwhile, this system also lacks granularity, providing little actionable insight into specific areas that may require improvement for businesses.

### **Problem Definition**

Recognizing these challenges, we propose a comprehensive solution focusing on incorporating time and sentiment analysis into the mix. We will provide a dynamic visualization that captures the nuanced shifts in customer sentiment over time. We will also utilize time series forecasting and polarity word clouds to provide insights into the anticipated performance and customer experience of a business, and thereby providing multifaceted insights beyond conventional star ratings. This solution aims to foster an informed consumer base and offer businesses valuable insights for continuous improvement.

### **Literature Survey**

To highlight the use of sentiment analysis in extracting insights and enhancing business performance, Luca (2016) examined how Yelp reviews impact restaurant revenue, revealing that a one-star increase can boost revenue by at least five percent. However, Luca relied solely on Yelp's rounded average rating system, while our project aims to measure consumer rating trends and thus offer more insight to prospective consumers than a mean star rating. Kauffmann et al. (2019) approached sentiment analysis by examining customer preferences at document, sentence, and aspect levels, aligning with our goal to identify business weaknesses based on customer preferences. While their analysis lacked trend consideration, we believe trend analysis is essential for suggesting business improvements. Beiji et al. (2016) addressed sentiment analysis on social media, sharing similarities with our project in aspect-level sentiment analysis and geographic data incorporation.

Various papers discuss the application of models to predict business ratings and cluster reviews. Li and Zhang (2014) used machine learning to predict restaurant star ratings and recommend restaurants based on reviews, aligning with our goal to incorporate customer sentiments into Yelp ratings predictions. Pang and Lee (2015) attempted numerical rating on a scale but only distinguished between positive, neutral, and negative sentiments. Turney (2002) categorized reviews as 'recommended' or 'not recommended' using unsupervised learning, serving as an initial step for our analysis, but lacked scaled assessment and trend analysis. Halde et al. (2022) employed machine learning and word clouds for sentiment analysis on Amazon product reviews, akin to our project. However, word clouds' effectiveness depends on data pre-processing and model selection, potentially offering less actionable insights. Dharaia et al. (2020) proposed polarity word clouds for various product categories, providing a distinct visualization of specific features' perceptions, surpassing traditional word clouds in emphasizing frequency. It's worth noting that most of the studies are conducted on data sampled at a specific point in time, and we believe there is room for improvement by factoring in time in analysis.

While there is a scarcity of literature directly linking time series analysis with sentiment scores, the potential enlightenment it can offer to businesses through time series forecasting is evident. Several literature also underscores the existence of issues in time-series analysis that warrant attention. Ching and de Dios Bulos (2019) emphasized leveraging social media and online reviews for valuable customer

insights and improving loyalty through text data analysis. They employed time series forecasting and linear regression in restaurant performance, but limitations on the reliability of linear regression for time series forecasting exist due to sparse daily reviews. Jha et al. (2015) tackle retail sales forecasting using a similarity heuristic for inventory clustering where historical sales data is sparse. This offers motivation to relate similar Yelp businesses with little review history to other similar, yet mature, businesses to develop insights. It is worth noting that issues may arise when adapting this approach because the metrics for determining similarity are not the same between businesses and inventory items. Harri and Brorsen (2009) explore using overlapping or unevenly spaced time series data and recommend the best estimators situationally. Their research points out that “missing observations can be a reason to use overlapping data,” addressing a sparsity issue in earlier observation periods. However, the results did not demonstrate any advantage in using overlapping data regarding prediction accuracy for multi-period forecasting.

### **Methodology**

Our project’s sentiment analysis approach offers a comprehensive, adaptable, and insightful solution that sets it apart from conventional sentiment analysis methods. In most of the literature we have reviewed, data is treated as static and does not account for dynamics in the forecasting and classification models, while our method incorporates a temporal dimension by dividing the dataset into half-year intervals. This temporal analysis enables a nuanced understanding of evolving sentiment trends over time, a feature often absent in traditional sentiment analysis. Additionally, our approach effectively mitigates data sparsity, a common issue found in daily data. By aggregating sentiment scores into extended timeframes, we circumvent the challenge of sparse or missing data points, thereby producing more reliable and accurate results. Furthermore, our project extends beyond sentiment analysis by incorporating predictive modeling, specifically the ARIMA model, to forecast sentiment scores. This time series-based temporal prediction capability sets our approach apart from conventional sentiment analysis, which often provides insights retrospectively. Furthermore, the project places emphasis on user interface integration. The use of Tableau for data visualization and the consideration of user interface aspects ensure that results are not only accurate, but also accessible and actionable for end-users. Lastly, our method is designed with practical scalability in mind which is rare for traditional sentiment analysis. Initially targeted at specific states, it can easily expand to any location, demonstrating suitability for broader applications.

In gathering essential data for our project, we leveraged the Yelp Open Dataset, encompassing both business-related information and customer review data. Following data retrieval from the Yelp API (Application Programming Interface), we processed it using AWS Athena, including the process of 1) merging two JSON files from the Yelp API: the `business` file containing essential business details such as business IDs, name, longitude, and latitude, and the `reviews` file, which contained business IDs and user reviews, and 2) conducting the sentiment analysis on all 10 million reviews. We initially applied AFINN sentiment analysis to assign sentiment scores to individual reviews due to its simplicity and ease of implementation, especially given the considerable size of the dataset and resource constraints. However, the dataset was still too large to handle for prediction within the scope of this project. Thus, we filtered for selected states to make processing more manageable and less computationally expensive. We focused on businesses from the three states that constitute a majority of the Yelp reviews dataset: Pennsylvania, Florida, and California. In the evolution of our sentiment analysis process, we transitioned from AFINN to a more sophisticated VADER scoring system. Introduced by Hutto and Gilbert in 2014, VADER proved to be a rule-based model consistently outperforming other sentiment analysis lexicons in the context of social media and reviews. VADER offers greater accuracy and understanding of sentiment,

accounting for factors such as intensity, degree of positivity or negativity, and negations. This shift aimed to improve sentiment analysis precision for a more refined evaluation of text in the reviews dataset.

Data standardization and cleaning were critical, especially when dealing with inconsistent city names. We made multiple attempts to leverage geocoding for obtaining accurate state and city names based on coordinates. However, taking Pennsylvania as an example, we had roughly one million data points, which would have necessitated 150 hours for processing given our resources. Consequently, we opted for OpenRefine to tackle inconsistencies, including misspellings and variations in capitalization. This involved applying clustering techniques to approximate accurate city names based on coordinates. An example of the cleaned dataset is shown below, followed by the summary statistics of all three states.

| business_id            | business_name                          | city          | state | latitude      | longitude       | stars | total_review_count | review_id              | rating | date                | review                                                                                | doc_sentiment |
|------------------------|----------------------------------------|---------------|-------|---------------|-----------------|-------|--------------------|------------------------|--------|---------------------|---------------------------------------------------------------------------------------|---------------|
| U3grYFleuRgAAQgdrHnw   | Brophy Bros - Santa Barbara            | Santa Barbara | CA    | 34.4037593    | -119.6939922    | 4.0   | 2940               | 7Dc0T0z0hKABobGhadQ    | 5.0    | 2005-03-01 16:57:17 | Words cannot describe how much I miss this place. The deck is right on the v          | 0.9807        |
| Aes-QQ_guDeYewMapFs_vg | Longboard's Grill                      | Santa Barbara | CA    | 34.410291545  | -119.6865421917 | 3.0   | 516                | WC0q5vNQiQkLK05KEs-vYQ | 2.0    | 2005-03-01 16:59:37 | Food is decent but is a complete tourist trap. Pretty spendy for the type of foo      | 0.8573        |
| CE0TnF8KilmJnBgkSDwA   | Pascucci                               | Santa Barbara | CA    | 34.419107     | -119.699243     | 3.5   | 1069               | Q0GJ06L78NkYnHr8dRlg   | 3.0    | 2005-03-01 17:25:13 | This place gets a 3.5...great location, great prices, good italian food after a day   | 0.722         |
| 29YqJwOGUeAWgH2Mc10A   | Paradise Cafe                          | Santa Barbara | CA    | 34.4200355    | -119.6968511    | 3.5   | 290                | TY9H-CAeqJkLn98KGuA    | 5.0    | 2005-03-01 17:47:15 | Lovely little restaurant which was converted from a house...lots of history to th     | 0.9328        |
| 6RBZlrmzE4NahJ7n1UPNA  | La Super-Rica Taqueria                 | Santa Barbara | CA    | 34.4279428    | -119.6872838    | 4.0   | 1759               | b5WJQAwKwYQcOGS-eCA    | 4.0    | 2005-03-01 17:59:26 | This little hole in the wall taqueria is written about in all the travel magazines. i | 0.2289        |
| jsuubXBnkSjIhbkUDmpQ   | Sushi Teri House                       | Santa Barbara | CA    | 34.4185472    | -119.7059044    | 3.5   | 203                | hX0XEPQ0AtUghHEQWQKQ   | 4.0    | 2005-03-01 19:33:35 | This is the original location for this great teriyaki house (the other two are in Gc  | 0.9179        |
| b5ID6RkUrodsA1QwngDg   | Wildcat Lounge                         | Santa Barbara | CA    | 34.4181393671 | -119.6984343008 | 3.5   | 288                | sg2Mc_sanoqZ-_D0XZkHqA | 5.0    | 2005-03-01 19:35:44 | Oh the Shitty Kitty...by far the best bar in Santa Barbara...Very retro, very viny    | 0.9166        |
| 8ASLSwsKK5vXDH1M3_T11w | The Ritz-Carlton Bacara, Santa Barbara | Santa Barbara | CA    | 34.4345451317 | -119.9206727994 | 3.5   | 1383               | zwJCV1vhtBQJyTLUcWbQ   | 5.0    | 2005-03-21 00:11:48 | Warning: so fantastic, she may think you are proposing! The premier spa/ressor        | 0.6652        |
| U3grYFleuRgAAQgdrHnw   | Brophy Bros - Santa Barbara            | Santa Barbara | CA    | 34.4037593    | -119.6939922    | 4.0   | 2940               | 70wYVJaHe7uF_YO383aXA  | 4.0    | 2005-03-21 00:20:59 | Casual seafood restaurant downtown on the water popular with the locals. Sti          | 0.5574        |
| yydsuHRIjgR3rwsN5vstA  | Sandpiper Golf Club                    | Goleta        | CA    | 34.4307328982 | -119.9072168552 | 3.5   | 85                 | h-uotwFATUD4C1NqXvKg   | 5.0    | 2005-03-21 00:39:01 | Sweet golf course right on the coast in Goleta, 15 minutes north of Santa Bart        | 0.4588        |
| DqVpNtgrPCP7r8rth-LREA | El Paseo Mexican Restaurant            | Santa Barbara | CA    | 34.4207686737 | -119.6991676593 | 3.5   | 282                | 8thvACH83JUC0KJJB5ng   | 4.0    | 2005-03-22 20:43:35 | This restaurant has got GREAT Mexican food and on a nice day the roof open            | 0.95          |
| AmAc0eD8lyy67QdQd01Mg  | Soho Restaurant & Music Club           | Santa Barbara | CA    | 34.4237983504 | -119.705293234  | 3.5   | 138                | e6NPv5weI8h8dLnMkyZQ   | 4.0    | 2005-03-22 20:45:10 | Soho is a great place to get a drink and watch some live music. There is a nice       | 0.906         |
| yE8RlXpIgr4XvRivkLw    | Chef Karim's Restaurant                | Santa Barbara | CA    | 34.423621     | -119.705492     | 4.0   | 28                 | Ckcb72WuB8Nds42JvcH-A  | 4.0    | 2005-03-22 20:46:09 | YUMMY Moroccan food here complete with belly dancers who get the whole                | 0.934         |
| edJoBss6n5F0BYh6pATTg  | The Palace Grill                       | Santa Barbara | CA    | 34.4179958952 | -119.6963100466 | 4.5   | 1500               | qnLJ-MDGL7N2CP7vLumw   | 4.0    | 2005-03-22 20:48:23 | This place is one of a kind. For great Cajun food this is the place to go in Sant     | 0.9601        |
| wrtz46YNQcQsU7Uil5w    | Chad's                                 | Santa Barbara | CA    | 34.41716      | -119.6990213    | 3.5   | 24                 | Eb5uJAFEK_yMR9vWEnd01w | 5.0    | 2005-03-22 20:50:40 | This is one of my favorite places in Santa Barbara. Very romantic and a great t       | 0.961         |
| Un-UD0vUv1co3TMMGaumA  | Shoreline Beach Cafe                   | Santa Barbara | CA    | 34.4028017    | -119.697943     | 4.0   | 534                | DEK-eTICQL-07SEEm7_PDQ | 5.0    | 2005-03-22 23:14:59 | You CANNOT miss out on this place for breakfast! There are tables in the sanc         | 0.9547        |
| 8ASLSwsKK5vXDH1M3_T11w | The Ritz-Carlton Bacara, Santa Barbara | Santa Barbara | CA    | 34.4345451317 | -119.9206727994 | 3.5   | 1383               | Lit-K9qNkY2IcRu8v4A    | 5.0    | 2005-03-29 19:29:36 | Bacara is amazing and makes for an indulgent romantic getaway. It feels secl          | 0.9076        |

Table 1. A Snapshot of the Dataset

|              | Size on Disk | Business # | Total Record # | Sentiment Score Range | Date Range      |
|--------------|--------------|------------|----------------|-----------------------|-----------------|
| Pennsylvania | 1.20 GB      | 31,638     | 1.59*1e6       | [-0.99, 0.99]         | 2005/2 - 2022/1 |
| Florida      | 0.82 GB      | 24,017     | 1.16*1e6       | [-0.99, 0.99]         | 2005/3 - 2022/2 |
| California   | 0.25 GB      | 5,203      | 3.45*1e5       | [-0.99, 0.99]         | 2005/3 - 2022/1 |

Table 2. Characteristics of Datasets for the Selected States (PA, FL, and CA)

The consolidation of sentiment scores, derived from individual reviews for each business, was initially conducted to enable the computation of a daily average sentiment score. Recognizing the uneven data sampling in the daily dataset, we strategically opted to aggregate sentiment scores into more balanced half-year intervals. In our pursuit of sentiment score prediction, we initiated the application of the AutoRegressive Integrated Moving Average (ARIMA) model for individual businesses. Each business was partitioned into an 80/20 train-test split where the chronological criterion guided the division. Throughout the ARIMA modeling process, we leveraged the `auto_arima` function to fine-tune hyperparameters associated with both trend ( $c$ ,  $t$ ,  $ct$ ,  $n$ ) and order ( $p$ ,  $q$ ,  $d$ ) for each individual business. The Mean Squared Error served as the metric of choice for evaluating the accuracy of predictions on the test datasets and hyperparameters were optimized to achieve the lowest MSE. After the optimization phase, all available data for each business was employed to forecast its performance for the subsequent half-year. A demonstration of ARIMA model fitting for a sample business is shown below, including training-data fitting and test-data prediction.



Figure 1. ARIMA Model Fitting for Training and Test Data: Demonstrating with a Sample Business

We opted to utilize Tableau as the primary framework for deploying our visualizations. However, Tableau falls short in advanced analysis, even with features like calculated fields and TabPy. In our efforts to enhance visualization and analysis, we decided to proceed by incorporating additional pre-processed data (ARIMA predictions, Vader Score, Aspect Level Sentiments) into the Tableau data source. We also incorporated the average sentiment score for businesses of the same type within the same city and compared it with the target business to provide further insights. To enrich our visualizations, for each business we included an overall word cloud (document level based on word frequency) and polarity word clouds (aspect level scaled by Vader Score and frequency). To implement this, we set up a server using the Flask library in Python. The server took specified state and business parameters to generate visualizations. While our current implementation included only overall word clouds and polarity word clouds, individuals interested in adding further analysis or visualization can easily establish additional routing to complement their analyses. Upon specifying parameters (selected state, selected business), our server generated and returned the visualization through a web browser. Our initial plan was to include the visualization URL directly within Tableau, but Tableau currently lacks the capability to work with dynamic links within existing worksheets. We are confident that our current Tableau and Flask setup allows for easy lateral expansion, enabling anyone to deploy their interactive visualizations seamlessly.

Below in Figures 2-5 is a rundown on how the Tableau dashboard works. When the dashboard is opened, the user will see a map with highlighted states for selected states in this project (Figure 2). The user can then either filter to a state using the filter at the top, or simply click on any of the states on the map. Clicking a state will take the user to the next sheet in the dashboard (Figure 3) that is composed of 2 parts: 1) the map is a zoomed in map of the selected state showing each business as a colored dot using the business's latitude and longitude coordinates, and 2) a graph hovering on the map displays the document score throughout the years, the average score by business category within the city, and predicted score in 6 months. Clicking on the business on the map or using the filter at the top will display scores for the selected business. Hovering over any of the points on the map or points on the graph will display business information including business name, ID, date of review, average rating, average stars, score, and review count. The filters at the top allows users to modify the selected state or search for a specific business by name, provided the user possesses this info. Clicking on the arrow labeled "Back to Map" will take the user back to the main map with highlighted states. Clicking the arrow labeled "Polarity Word Cloud" will take the user to a new page in their browser depicting the polarized word clouds for the selected business (Figure 4). Polarity word clouds include negative and positive word clouds at the people, product, and organization levels. Clicking the arrow labeled "Document Word Cloud" will take the user to a new browser page depicting just one overall word cloud for the selected business (Figure 5).

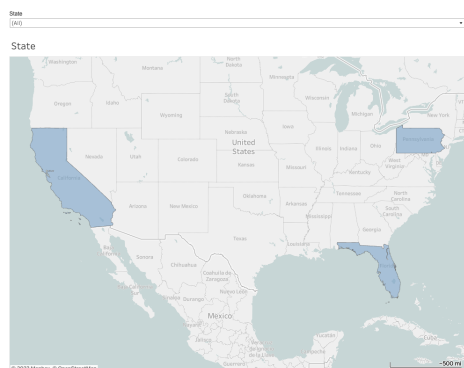


Figure 2: Tableau Dashboard Main Page

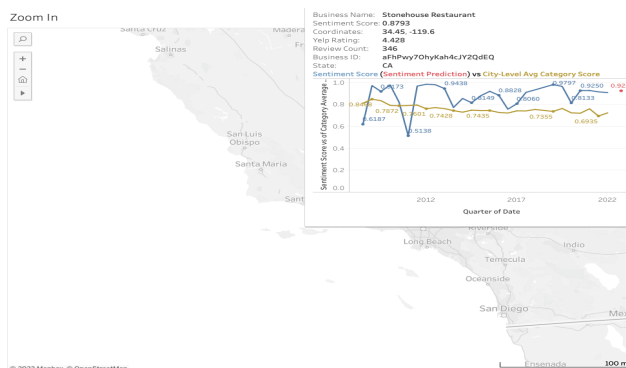
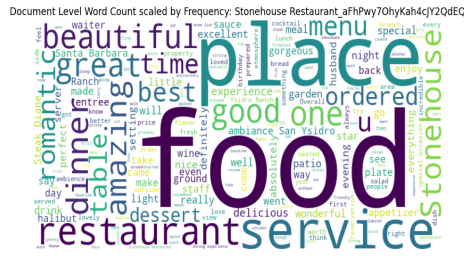


Figure 3: Visual Representation of Sample Business

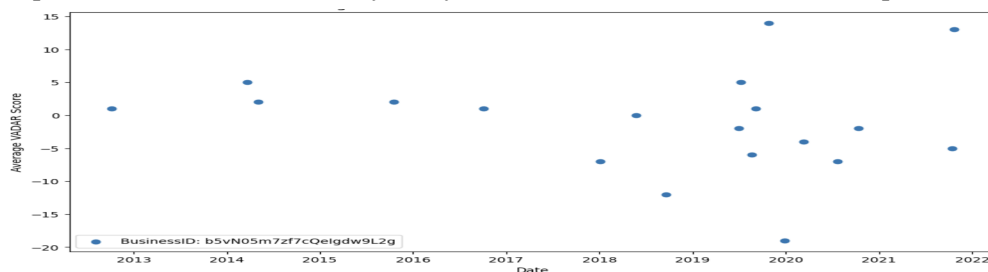


For the sample business, their performance stands out with high scores and surpassing the city average throughout the years. Nonetheless, a concern emerges as the average score for similar businesses in the city is on a downward trajectory. This trend suggests an unexplored and potentially relevant concern that could affect similar businesses within this city, and it warrants attention from both potential customers and the business owner. The polarity word clouds offered further insights. Notably, Laurent Perrier/Foxen/Jidori received consistent praise, while Dave Miner and Alfafa draw substantial criticism.

To address the issue of missing data (e.g. no reviews for over six months), we used a linear interpolation from the prior and subsequent time periods where data is available. If review data is still too sparse, that business was filtered out. To ensure the improvement in our model’s predictive accuracy is not the result of overfitting, we divided the data of each business using a chronological 80/20 train-test split and measured the predictive accuracy using mean squared error. Leveraging the model generated based on training data across all three states, the mean square errors for test data are consistently low—specifically, 0.22 for PA, 0.23 for FL, and 0.17 for CA. This achievement is particularly noteworthy given the substantial size of the dataset, exceeding 3 million records. Total number of business, number of business forecasted, and MSE for the three selected states are shown below.

|              | Total Business # | Forecasted Business # | Test MSE |
|--------------|------------------|-----------------------|----------|
| Pennsylvania | 31,638           | 29,237                | 0.22     |
| Florida      | 24,017           | 21,704                | 0.23     |
| California   | 5,203            | 4,801                 | 0.17     |

We considered whether businesses may benefit from a short-term prediction of user reviews, e.g. “What do the next 15 customers think about the business?” in addition to a forecast of sentiment score for the next six-month period. However, our analysis revealed that review data for many businesses were highly sparse and thus predictions of the next several review sentiments would not be accurate and effective for business insights overall. As depicted below, this particular business exhibits a sparse review frequency, with only 1-5 reviews per year. This minimal data density proves challenging for accurate short-term predictions, and unfortunately many businesses in the dataset share the sparse data pattern.



5

To optimize user interactivity, we conducted thorough test runs both within our team and by involving friends and family members in the evaluation process. Eventually, we decided to transition from TabPy to Flask. Although our initial attempt with TabPy for executing Python scripts to generate word clouds showed some success—producing word clouds when clicking on businesses in the Tableau map—several challenges surfaced during user testing: 1) Complex setup process: the overall setup using TabPy was not straightforward as users were required to establish a local server, presenting a considerable challenge for those without extensive experience in this domain; 2) Unsatisfactory computation time: generating word clouds takes over 20 seconds per business on TabPy, which is inconvenient for users during the interactive experience; 3) Limited functionality: Tabpy was merely equipped to return string/binary/int/float, but not advanced analysis nor additional charts; and 4) Limited accessibility: we discovered that TabPy is unavailable on Tableau Public, thus utility is restricted to a local environment, requiring users to download the Tableau workbook, associated data files, and operate on their local machines. Given these challenges and the feedback received, we opted to explore alternatives. Although server setup is still needed, Flask emerged as a more viable solution over TabPy for reduced computation time, more powerful functionality, and improved accessibility. This transition addressed the aforementioned issues and also contributed to an overall improvement in user experience and efficiency.

To evaluate the scalability of our analysis and interactive tool, we initially designed our solution to focus exclusively on businesses within Pennsylvania. We applied filters, considering factors such as city or minimum review count. Subsequently, we expanded our criteria to encompass all businesses available within the states of Pennsylvania/Florida/California. Assuming that the data for businesses in other states follows a similar format, as we believe it does, we can easily incorporate additional states or even data from across the United States. This approach allows us to include the outputs of sentiment predictions for businesses in these regions within our visualization.

### **Conclusion**

The project effectively addresses the limitations inherent in Yelp's current rating system, providing a more dynamic and intricate portrayal of consumer experiences. This project holds significant potential in two key dimensions: 1) Consumer empowerment: the project strives to empower consumers by offering a nuanced comprehension of businesses, transcending the constraints imposed by static star ratings, and 2) Business improvement: businesses stand to gain valuable insights through sentiment analysis and time series forecasting, enabling them to make informed enhancements.

However, several limitations persist in this project, including 1) Server setup requirement: the generation of word clouds is constrained by the establishment of a user-friendly server, which poses a potential hurdle for users who may find the server setup process less intuitive, and 2) Computational resources: the processing demands of the dataset led to downsizing by focusing on specific states.

Future directions for the project include: 1) Geographical expansion: extend the analysis to encompass businesses in additional states and regions for a more comprehensive understanding, 2) Real-time analysis: implement real-time sentiment analysis to capture the most current consumer experiences, 3) Diversification of models: explore the integration of splines and reinforcement learning models beyond ARIMA for improved sentiment score prediction, 4) Enhanced Aspect-Level Analysis: explore and integrate new tools for a more intuitive representation of sentiments associated with specific aspects of businesses, and 5) Enhanced accessibility: establishing a cloud server eliminates the necessity for a local server and improving overall efficiency.

All team members have contributed a similar amount of effort.



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